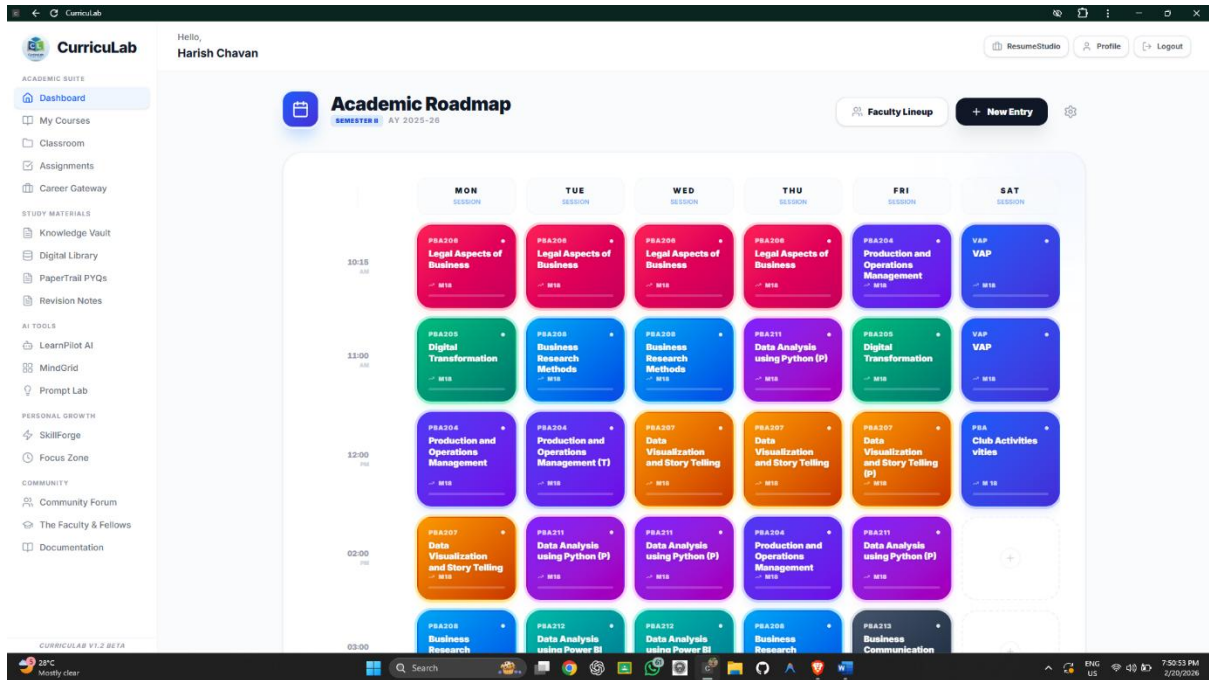


CurricuLab

The Student Centric Study Management System



The Technical Team

Harish Ravindra Chavan

Code Base Manager and Core Developer

Kaustubh Nandurkar

APIs and Database Manager

Version 1.2 beta

Team Details

Harish Ravindra Chavan is the **Student of MBABA, Sem-2 from Sandip University** and for this project he is **Code Base Manager and Core Developer**, responsible for creating a frontend in order to convert the idea into the real product that fulfilling the all student's expectations with easy to handle UI design contributing user friendly experience.

However, **Kaustubh Khushal Nandurkar** is also the **Student of MBABA, Sem-2 from Sandip University** and he is the **APIs and Database Manager** for this project, responsible for collecting different free API keys and connecting the frontend to the Backend using the database named 'Supabase' through its keys and SQL scripts(in order to create different Tables and Buckets).

Acknowledgement

The Special Thanks to the our teachers Dr. Zahir Shaikh and Dr. Samadhan Bundhe from Sandip University for their precious guidance and support. We are glad to have such guide or mentor who help us to review and evaluate the process continuously to avoid any neglecton.

Under the Guidance of Dr. Zahir Sir, we have applied the various concepts of Business Research Methods, while under the Guidance of Dr. Samadhan Bundhe, we have applied the concept of Data Management and Warehousing along with SQL handling.

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Vision And Purpose

Technical Team's vision is to create a platform that is reliable, scalable, and easy for students to use, and that brings together a number of academic tools like ChatGPT, Gemini and Google Classroom into one system. The goal is to make sure that students can easily access study materials, announcements, assignments, and AI tools without having to switch between different apps.

Key Purpose are:

- Turning approved content into system modules that can be used
- Keeping the system stable and running well
- Making sure that data is stored and retrieved safely
- Helping with ongoing updates and improvements
- Making sure that users have a smooth experience on all devices

Technical Requirements

To construct and use the CurricuLab system, we need simple gear, software, and an internet connection.

(1) **Hardware:**

To develop the system, a computer or laptop with at least 8 GB RAM, a modern processor, enough storage, and a stable internet connection is needed.

Students can use their phones, tablets, laptops, or desktops to access the platform from anywhere.

(2) **Software:**

The platform works on Windows, macOS, and Linux and runs through any modern web browser. For development, tools such as an IDE, version control (Git), and the Node.js environment are used to build and manage the application.

(3) **Database and Backend:**

Supabase is used to manage the database, user authentication, and storage. It helps keep student data and academic resources organized, secure, and easy to access.

(4) **Web Technologies:**

The system is built using Next.js to create a fast and scalable web application, while Tailwind CSS is used to design a clean, responsive interface that works well on all screen sizes.

(5) **API Integration:**

The platform connects with external services through APIs. For example, Google Classroom integration allows assignments and study materials to be accessed in one place, and AI providers support smart tools within the system.

(6) **Security and Performance:**

Basic security measures such as user login, safe data handling, and controlled access are included to limit users and ensure security.

The system is also optimized to load quickly and work smoothly even when many students use it at the same time.

Technical Workflow

Step-1 Requirement Analysis

First, We Understood system needs & Expectations from the Coordinator and other teams. It helps us to define detailed Project Requirements

Step-2 System Design

Using Project Requirements, we planned architecture, database schema, and module structure after the series of discussion with R&D Manager(Akash Jayaprakash Mangalamthodi)

Step-3 Development

We Build frontend interfaces and backend logic that creates seamless user experience using Available tools and technologies.

Step-4 Integration

As we have frontend ready, we connected APIs, databases, and external services to it to enhance overall user experience.

Step-5 Testing

During the development, we required continuous review of functionality, usability, and performance and here our Quality Assessment Manager (Kaif Shah) helped us to achieve project expectation.

Step-6 Deployment

Next, we published or deployed validated UI with content and features using GitHub and Render with free tier in order to save money and other resources.

Step-7 Maintenance

After Deployment, we made so many change with bug fixes, module updates, and workflow improvements in order to enhance overall performance.

Collaboration with Other Teams

The Technical Team worked closely with all departments. From **Coordinator**, we received project direction, priorities, and workflow alignment. **Data Acquisition Team** hand overed study materials in structured formats (HTML files) to ensure quality and consistency. However **Research & Quality Team** helps us to validated content to ensure academic accuracy and with their continuous reviews and suggestion, we converted Idea into the fully functional system.

This collaboration ensures that the final platform is both technically sound and academically reliable.

Team	Role	Name	Responsibility
Coordinator Team	Workflow & Project Manager (Coordinator)	Tanu Chaudhary	Leads the project, defines requirements, coordinates all teams, monitors progress, and ensures timely completion
Data Acquisition Team	Resource Manager & Team Leader	Anukriti Srivastava	Manages resources, guides members, oversees content creation, and supervises subject allocation
Data Acquisition Team	Assistant Resource Manager	Tejas Patil	Assists in data collection, resource handling, and preparation of study materials
Data Acquisition Team	Junior Assistant Manager	Georgetta Wilson	Handles assigned subjects and supports content development
Data Acquisition Team	Junior Assistant Manager	Manishankar Veermalla	Assists in subject-wise content creation and resource organization
Research & Quality Team	Quality Assessment Manager	Kaif Shah	Reviews materials for accuracy, syllabus alignment, formatting, and final approval
Research & Quality Team	R&D Manager	Akash Jayaprakash Mangalamthodi	Improves content generation methods, evaluates AI outputs, and supports quality enhancement

Tools and Technologies Used

We used the following Tools and Technologies at different stages of the project where needed:

- Antigravity IDE for frontend generation
- Supabase for backend database and storage
- External AI APIs for intelligent features
- Google Classroom API for academic integration
- Web technologies for interface design

Smart AI Integration for Reliable Performance

During the development of CurricuLab, multiple AI providers were integrated to LearnPilot, Revision Notes and Assignments. Free-tier AI services often have limits such as usage caps, slow responses, or temporary outages once quotas are reached, which can affect system reliability during busy academic periods.

To avoid disruptions without additional costs, a smart auto-switching system was implemented. The platform normally uses a primary AI model, but if limits are reached or performance drops, requests are automatically redirected to other available models. This ensures continuous service, reduces downtime, balances usage across providers, and keeps the system stable and cost-effective.

Learning Outcomes

This project provided us a hands-on experience in:

- Full-stack web development
- API integration
- Database management
- Cloud-based backend services
- Real-world software project workflow
- Collaborative development practices

Conclusion

The Technical Team serves as the operational core of CurricuLab. By converting ideas, validated content, and requirements into a robust digital platform, the team ensures that students receive a seamless, organized, and reliable academic management system.

It enables the entire project to function as a unified solution rather than a collection of disconnected resources.